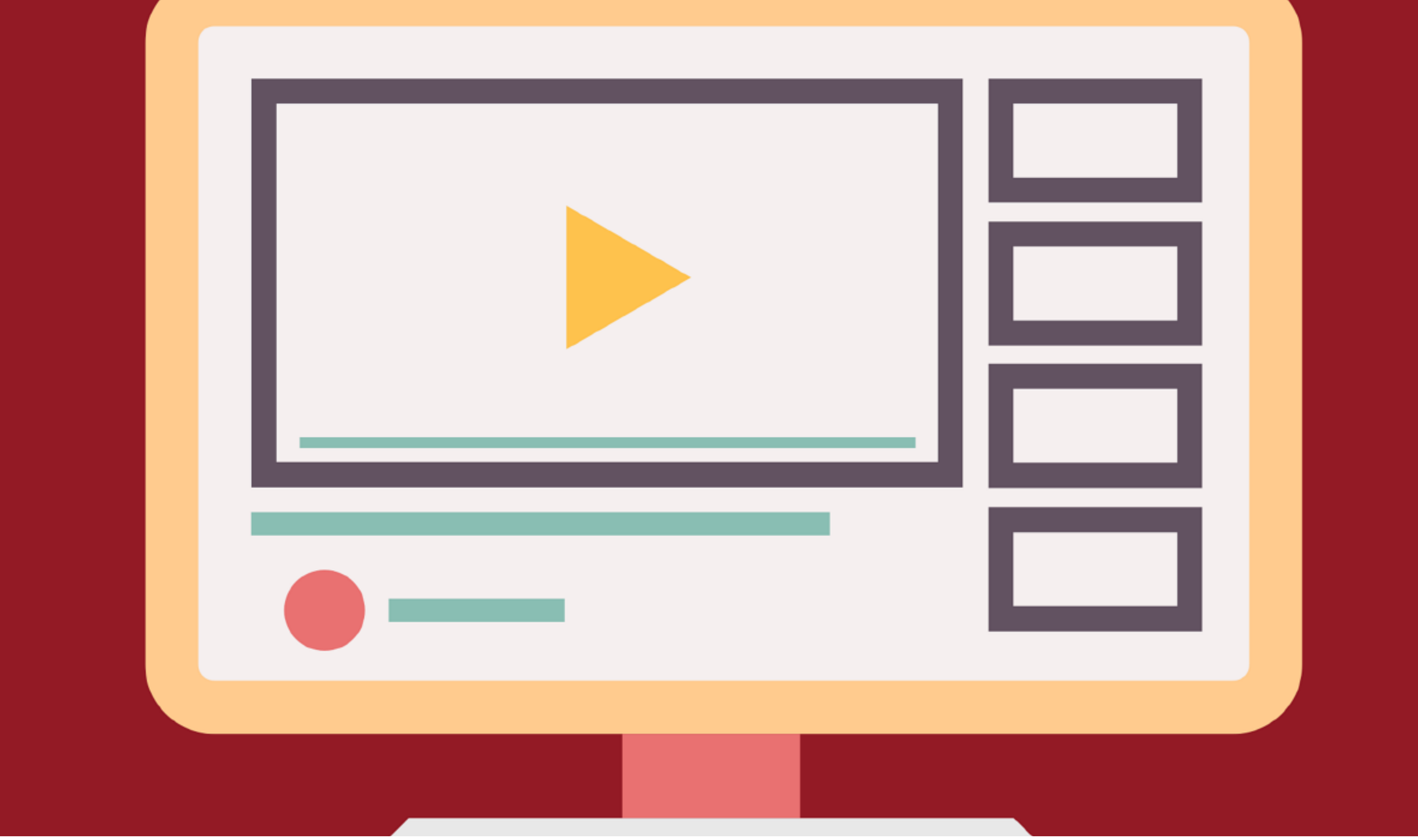

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Elastic Constants of β-$(Al_xGa_{1-x})_2O_3$ for Heterogeneous Interface Applications

The elastic and piezoelectric properties of β-$(Al_xGa_{1-x})_2O_3$, with compositions ranging from $x = 0.125$ to 0.25 were analyzed using first principles calculations. The controllable bandgap by Al concentration make it a promising gate dielectric material for Ga_2O_3-based high electron mobility transistors (HEMTs). The computed 13 monoclinic elastic constants reveal composition-dependent trends in mechanical and electrical behavior, underscoring the influence of aluminum concentration on structural stability. Composition-dependent trends in the 13 monoclinic elastic constants are presented from first-principles for

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$x = 0 - 0.25$, without prescribing a specific analytic form; these trends inform the design of strain-engineered materials for advanced semiconductor applications. These findings provide valuable insights into β-$(Al_xGa_{1-x})_2O_3$ and pave the way for improved performance in next-generation Ga_2O_3 HEMTs.

In the alloy modeling, for a certain elemental ratio, the alloy configuration with the maximum inter-Al distances were chosen for the analysis among various possible configurations. This ensures a systematic and representative selection of configurations for DFT simulations. For alloy modeling, the host β -Ga_2O_3 crystal was taken from the Materials Project (material ID mp-886; β-phase, C2/m).

I. INTRODUCTION

In Power electronics, significant advancement has been made with various wide-bandgap semiconductors, including SiC, GaN, and InP [1]. In conjunction with these materials, monoclinic(β-phase, space group C2/m) β-Ga_2O_3 has emerged as a promising candidate due to its large bandgap (~4.8 eV), high breakdown field (~8 MV/cm), and high electron mobility (~300 cm^2/V.s) [2]. Here the monoclinic β-angle is defined as the angle between a and c $(\beta \neq 90^\circ)$. Despite these advantages, the demand for more versatile materials has prompted research into its alloys, particularly β-$(Al_xGa_{1-x})_2O_3$, where Al substitutes for Ga in β-Ga_2O_3 .

Recent advances in $(Al_xGa_{1-x})_2O_3$ thin-film fabrication have further underscored its stability and applicability for power electronics and optoelectronics [3]. In particular, the controlled variation of Al content in β-$(Al_xGa_{1-x})_2O_3$ allows a fine-tuning of lattice parameters and bandgap, which can be utilized to enhance the device performances and stability in devices where lattice mismatch is unavoidable [4]. Especially, β-$(Al_xGa_{1-x})_2O_3$ has received considerable attention for high-power applications for the high electron mobility transistors (HEMTs). This alloy's potential stems from its tunable structural and electrical properties, which can be modified by controlling the Al content [5]. The introduction of Al into the Ga_2O_3 on Ga_2O_3 substrates creates a lattice mismatch that induces strain, which then create electric field due to its strong piezoelectricity. These changes significantly alter the material's electronic structure, affecting energy bands and levels in ways similar to those observed in GaAs and AlGaAs or GaN and AlGaN [6], [7].

The inherent piezoelectricity of β-Ga_2O_3 is amplified in β-$(Al_xGa_{1-x})_2O_3$ due to the stronger polarization by Al addition, resulting in increased piezoelectricity [8]. This property makes β-$(Al_xGa_{1-x})_2O_3$ suitable for energy devices that rely on piezoelectric responses. However, to fully utilize these properties, a thorough understanding of the material's elastic properties is essential, as they determine how the material deforms under mechanical strain and directly influence its piezoelectric response [9]. Piezoelectric coefficients and elastic constants are directly related, as the piezoelectric effect depends on the interaction between mechanical strain and electric polarization. Elastic constants define a material's mechanical stiffness, while piezoelectric coefficients quantify its ability to generate electric charge under stress. Understanding this relationship is essential for modeling β-$(Al_xGa_{1-x})_2O_3$'s electromechanical properties in piezoelectric applications [10]. Specifically, the electrical charge generated when the material in strained depends on the elastic compliance coefficients.

In this study, we investigate the elastic constants of β-$(Al_xGa_{1-x})_2O_3$ based on density functional theory (DFT) and discuss piezoelectric properties using density functional perturbation theory (DFPT) values [11]. The challenge in modeling an alloy, however, is addressing the randomness in the elemental distribution properly. A simplified yet realistic model for the DFT analysis is devised to address this issue.

The remainder of this paper is organized as follows: Section II presents the β-$(Al_xGa_{1-x})_2O_3$ alloy modeling. Section III details the piezoelectricity analysis (literature DFPT e combined with our DFT-calculated C to estimate $d = e \cdot C^{-1}$). Finally, Section V summarizes our findings and discusses their implications for β-$(Al_xGa_{1-x})_2O_3$ based semiconductor applications.

II. THEORETICAL ANALYSIS

A. FIRST-PRINCIPLES CALCULATIONS OF β-$(AL_XGA_{1-X})_2O_3$ ALLOY

In the alloy modeling, for a certain elemental ratio, the alloy configuration with the maximum inter-Al distances were chosen for the analysis among various possible configurations. This ensures a systematic and representative selection of configurations for DFT simulations. For alloy modeling, the host β-Ga_2O_3 crystal was taken from the Materials Project (material ID mp-886; β-phase, C2/m). The conventional 20-atom cell was expanded to a $1 \times 1 \times 2$ monoclinic supercell (40 atoms: 16 cations and 24 O atoms) following the crystallographic dataset compiled by Jain et al. [25]. Two and four Ga atoms were then substituted by Al to construct

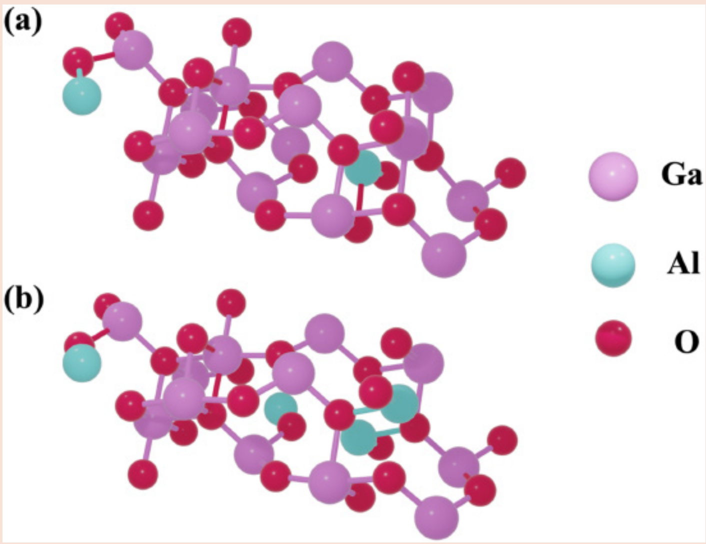

FIGURE 1 Comparative images of β-$(Al_xGa_{1-x})_2O_3$ samples with varying compositions: (a) x=0.125, indicating an aluminum concentration of 12.5%; and (b) x=0.25, with an aluminum concentration of 25%. Each image showcases the material's microstructure under these distinct compositional variations.

TABLE 1 **Volume, density, and beta angle comparison of β-$(Al_xGa_{1-x})_2O_3$ with varying Al composition in this study.**

AL COMPOSITION	VOLUME (Å^3)	DENSITY (g/cm^3)	β(°)
0	412.21	6.04	103.69
0.125	428.56	5.48	104.52
0.25	440.13	5.01	105.14
0.5	471.81	4.07	107.31
Quantity	**This work (DFT)**	**Experiment (x=0)**	**Δ(%)**
a (Å)	12.2241	12.21	0.115
b(Å)	3.0156	3.04	−0.803
c (Å)	5.7545	5.81	−0.955
β (deg)	103.6909	103.87	−0.172
V_{40} (Å^3)	412.21	418.74	−1.559
ρ (g·cm$^-$)	6.0409	5.9465	1.572
x comp.	**0**	**0.125**	**0.25**
β(this)[°]	103.69	104.52	105.14
β(exp.)[°]	103.87	103.939	104.008
Δβ(%)	−0.173	0.559	1.089
V_{40} (this)[Å3]	412.21	428.86	440.13
V_{40} (exp.)[Å3]	418.74	411.392	404.131
ΔV_{40} (%)	−1.559	4.246	8.908
ρ(this)[g·cm^{-3}]	6.04	5.48	5.01
ρ(exp.)[g·cm^{-3}]	5.947	5.708	5.459
Δρ(%)	1.572	−3.989	−8.225

x = 0.125 and x = 0.25 configurations, respectively(Figure 1).The remaining configurations were ranked based on total interatomic distances, and the structure with the lowest crystal energy were selected for further analysis.

All structures were geometry-optimized via variable-cell relaxation at zero external pressure using Quantum ESPRESSO [10] with local density approximation (LDA) norm-conserving pseudopotentials. The wave function cut-off energy of 230 Ry and the kinetic cut-off energy of 1380 Ry. The convergence threshold of the electronic energy was set to 1.0×10^{-10} Ry, and ionic force threshold was set to 2.0×10^{-3} *Ry/Bohr*. $4\times8\times8$ Monkhorst–Pack grid were used for the unit cell k-point sampling, and a comparable grids were chosen for the 40-atom supercell supercell calculation. After vc-relax, the optimized models exhibited only minor angular deviations relative to prior reports [9]. Self-consistent electronic steps were considered to be converged when the total-energy change became smaller than $1\times10^{-10}Ry$ ($1.4eV\times10^{-9}$ per 40-atom cell). During ionic relaxation, the convergence criteria was when the maximum Hellmann–Feynman force dropped below 2×10^{-3}Ry/Bohr and the residual stress was less 0.01 kbar. Tightening these criteria by an order of magnitude ($1\times10^{-11}Ry$ for the electronic threshold and $2\times10^{-4}Ry\,/\,Bohr$ for forces) changes any elastic constant, C_{ij}, by less than 0.3 GPa, confirming numerical convergence. The incorporation of Al into the structure results in noticeable changes in volume, density, and beta (β) angle. Table 1 summarizes the trends in volume, density, and β with varying Al compositions. For $x=0$, our lattice is $a=12.2241$ $^-$ and $b=3.0156$ $Å$. Also $c=5.7545Å$, $\beta=103.6909^{\circ}$, $V_{40}=412.21$ $Å^3$, $\rho=6.0409$ $g\cdot cm^{-3}$. Deviations from Kranert [23] are *+0.12%, -0.80%, -0.96%, -0.17%, -1.56%, +1.57%*. These values are in good agreement with experiments, within $1-2\%$ overall (Table 1). For experimental comparison across composition, we adopt the $x=0$ intercepts from Kranert et al. and the composition slopes from Galązka et al. These two sources define the experiment-based references used in Table 1. [23], [24].

Intriguingly, as the Al composition increases, the volume slightly expands from $412\text{Å}^3\,(x=0)$ to $472\text{Å}^3\,(x=0.5)$, even though Ga was substituted with the smaller Al cations, representing the lattice distortions by stronger polarity. The β also increases from *103.69°* to *107.31°*, reflecting structural changes typical of monoclinic systems. The density decreases linearly from $6.04\,g/cm^3$ to $4.07\,g/cm^3$, consistent with the expected reduction in mass density as the lighter Al atoms replace Ga atoms. The simultaneous increase in β and volume suggests a coupling between angular distortions and lattice expansion, characteristic of monoclinic structures accommodating compositional changes. The observed nonlinear trends of volume and β, without assuming a specific polynomial form, likely arise from ionic size mismatch and the resulting anisotropic lattice distortions. These trends provide insights into the tunability of structural properties in β-$(Al_xGa_{1\text{-}x})_2O_3$ which could be advantageous for optimizing materials for electronic and optoelectronic applications.

For $x > 0$, our results agree with an experiment-based model (Kranert intercepts; Galazka slopes), see Table 1 (lower panel) [23], [24].

The consistency of these results with experimental data validates our alloy construction approach and its potential for predicting properties of other Al-Ga-O compositions with monoclinic structures.

B. ELASTIC STIFFNESS CONSTANT

Independent homogeneous strain modes were applied in Voigt notation $\varepsilon_1 \equiv \varepsilon_{xx}$, $\varepsilon_2 \equiv \varepsilon_{yy}$, $\varepsilon_3 \equiv \varepsilon_{zz}$, $\varepsilon_4 \equiv \varepsilon_{yz}$, $\varepsilon_5 \equiv \varepsilon_{xz}$, $\varepsilon_6 \equiv \varepsilon_{xy}$ with amplitudes from -0.5% to +0.5%. For each applied strain, the strained cell was kept fixed while internal atomic coordinates were relaxed prior to evaluating the total energy. The strain–energy data were then used to extract the elastic stiffness constants. The elastic stiffness constants, $C_{ij} \equiv c_{\alpha\beta\gamma\delta}$, are extracted as follows:

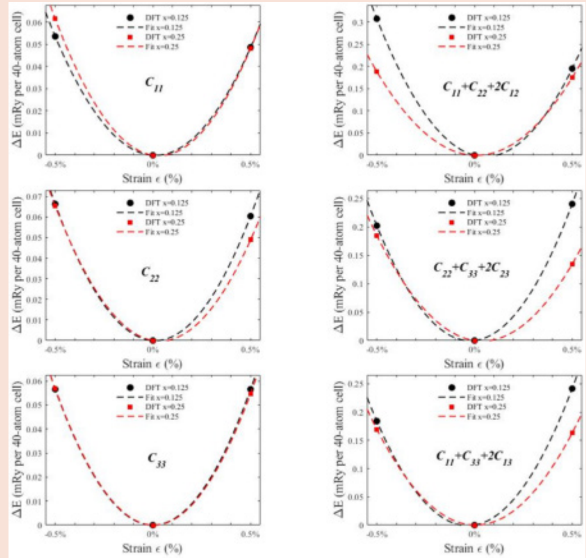

FIGURE 2 Strain vs. Total energy curves for calculating the elastic stiffness constants for the β-$(Al_xGa_{1-x})_2O_3$model. The curves were fitted to a second-degree polynomial, and the elastic constants were extracted from the $\frac{A_0}{\Omega}$ term.

TABLE 2 Calculated elastic constants with comparison to previous theoretical and experimental results.

ELASTIC CONSTANT(GPa)	GA_2O_3		$(AL_xGA_{1-x})_2O_3$	
	REF. [19] *(EXPT.)*	*THIS WORK*	$x = 0.125$	$x = 0.25$
C11	242.8	246.7	283.0	295.6
C22	343.8	350.4	351.0	308.3
C33	347.4	346.9	313.0	300.6
C12	128.3	154.3	171.4	141.6
C23	70.9	92.15	279.7	125.7
C13	160.0	167.2	167.0	150.5
C55	88.6	76.03	124.8	61.84
C15	-1.6	-2.898	-13.51	-14.26
C44	47.9	52.66	88.47	52.32
C66	104.0	109.02	479.1	150.6
C25	0.4	36.10	12.73	6.782
C35	1.0	28.47	2.875	13.82
C46	5.6	10.56	12.76	42.30
C16	0.0	23.92	41.11	20.73

$$C_{ij} \equiv c_{\alpha\beta\gamma\delta} = \frac{1}{\Omega} \cdot \frac{\partial^2 E_{total}}{\partial\epsilon_{\alpha\beta}\partial\epsilon_{\gamma\delta}} \tag{1}$$

where $\varepsilon_{\alpha\beta}$ is the strain tensor, and E_{total} is the total energy of the unit cell. The C_{ij} was calculated by applying independent strains and then fitting the results to energy-strain relations [11]. The stiffness constants, $C_{ij} \equiv c_{\alpha\beta\gamma\delta}$ are based on Voigt compound indices, where Ω stands for the unit cell volume [12].

The strain dependence of E_{total} was fitted to a second order polynomial:

$$E_{total}(\varepsilon) = \Omega U(\varepsilon) = A_0\,\varepsilon^2 + A_1\varepsilon + A_2 \tag{2}$$

TABLE 3 Local quadratic interpolant $C_{ij}(x)$ based on Al composition for β-$(Al_xGa_{1-x})_2O_3$.

ELASTIC CONSTANT	THREE-POINT LOCAL QUADRATIC INTERPOLANT $C_{ij}(X)$[GPA]
C11	$-758.4x^2 + 385.2x + 246.7$
C22	$-1386x^2 + 178.0x + 350.4$
C33	$688.0x^2 - 357.2x + 346.9$
C12	$-1501x^2 + 324.4x + 154.3$
C13	$-521.6x^2 + 63.60x + 167.2$
C23	$-1093x^2 + 2866x + 92.20$
C44	$-2304x^2 + 574.4x + 52.73$
C55	$-3577x^2 + 837.6x + 76.01$
C66	$-2236x^2 + 5755x + 109.0$
C15	$313.6x^2 - 124.0x - 2.927$
C25	$560.1x^2 - 257.2x + 36.14$
C35	$1168x^2 - 350.8x + 28.52$
C46	$876.8x^2 - 92.07x + 10.53$
C16	$-1203x^2 + 288.0x + 23.98$

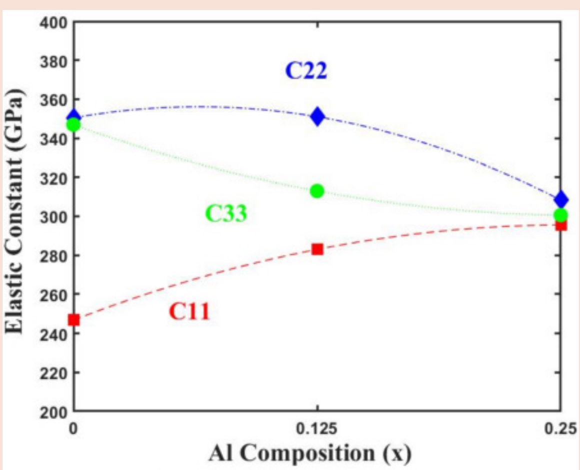

FIGURE 3 Normal strain (C_{11}, C_{22}, C_{33}).

In Figure 2, symbols are DFT total-energy differences $\Delta E(\varepsilon)$ at $\varepsilon = -0.5\%, 0\%, +0.5\%$ $(x = 0.125, 0.25)$. The curves are guides to the eye only, and C_{ij} are obtained from the curvature at $\varepsilon = 0$. Eq. (2) is provided only to illustrate the local $\varepsilon-$dependence of the total energy and the reported C_{ij} are determined from the symmetric three-point curvature at $\varepsilon = 0$ (central difference). The β, a defining feature of the monoclinic β-phase structure, deviates from 90° and turns out to be very sensitive to compositional and positional changes. This variation in β prompted further investigation into the atomic relaxation behavior, which revealed dependency on the specific positioning of the Al atoms [13]. During the atomic force relaxation process for the geometry optimization, this angle varied from *102°* to *104°*, reflecting the influence of different Al positioning. These findings provide insights into the structural dynamics of the β-$(Al_xGa_{1-x})_2O_3$ alloy, enhancing the understanding of how Al atom positioning affects the alloy's behavior.

III. RESULTS AND DISCUSSION

A. ELASTIC PROPERTIES OF β-$(AL_XGA_{1-X})_2O_3$

The elastic constants of β-$(Al_xGa_{1-x})_2O_3$ were evaluated at $x = 0, 0.125$ and 0.25; within this dilute range we discuss trends, including local bowing, directly from these DFT points using the discrete second difference $\Delta^2 C = C(0.25) - 2C(0.125) + C(0)$. These fitted equations capture the nonlinear changes in elastic constants as Al content varies, and provide the estimation of piezoelectricity for an arbitrary x between 0 to 0.5. Observations alone do not fully capture the complexity of these trends, making polynomial fitting a valuable tool for representing and predicting the material's elastic behavior. Table 3 lists three-point local quadratic interpolants that reproduce exactly the DFT values at $x = 0, 0.125$ and 0.25; they are provided for convenient interpolation within $x \le 0.25$ and are not used for inference or extrapolation. Curvature (bowing) is discussed locally from the three DFT points.

Each elastic constant responds differently to Al composition. For instance, C_{11} and C_{44} initially increase with the addition of Al, reflecting the stiffness of the crystal increases. At higher concentrations, these values plateau or decrease, demonstrating the dual effect of Al incorporation: initial stiffening followed by a strain-induced softening at higher compositions [14].

C_{22} decreases steadily, indicating progressive weakening of bonding strength along a specific crystallographic direction. In contrast, C_{33} follows a parabolic trend, decreasing initially due to lattice distortion before increasing again as the structure accommodates compositional changes [15]. Figure 3 visualizes these trends using the three DFT markers, and the smooth lines are visual connectors rather than fits or models.

As shown in Figure 4, the bonding strength in planar directions, represented by C_{12}, C_{13}, and C_{23}, generally decreases with the increasing Al contents except for C_{23}. C_{12} decreases consistently, indicating a reduction in bonding strength as smaller Al atoms replace larger Ga atoms. This linear decrease suggests a uniform softening effect across compositions. C_{13}, although only slight decreases was shown compared to C_{12}, indicates lower sensitivity to Al composition. This limited change may be the result of the fact that it's the planar strain constants is limited into a certain direction and affected less by Al substitution. Interestingly, C_{23} exhibits a parabolic trend, peaking up at intermediate Al compositions. This behavior reflects the influence of Al distribution and positioning, which create competing effects within the lattice. Figure 4 depicts a nonlinear composition dependence, illustrating how planar bonding interactions evolve with structural changes[16].

The shear strain resistance, represented by C_{44}, C_{55}, and C_{66} demonstrate how Al composition influences the lattice under shear deformation. As shown in Figure 5, C_{44} increases initially due to lattice stiffening effects but declines at higher Al compositions, reflecting structural weakening caused by the strain. Similarly, C_{55} exhibits a comparable trend, where resistance rises initially but decreases as the saturation effects dominate. C_{66} follows a nonlinear trajectory, rising sharply at lower compositions, whereas declines after a peak, indicating the complex lattice interactions under shear deformation [17].

C_{16}, theoretically expected to be zero due to symmetry, fluctuates significantly. Despite the symmetry expectations, C_{16} shows non-zero values due to local asymmetries introduced by non-uniform Al distribution within the lattice [18].

B. STRAIN-ENGINEERED INTERFACES IN β-GA_2O_3/ALGAO: IMPLICATIONS FOR HEMTs

The relevance of elastic constants and piezoelectric properties of β-$(Al_xGa_{1-x})_2O_3$ was examined in regards of strain management and mechanical stability for HEMT applications. In particular, the quantitative relationship between Al composition and the material's mechanical and piezoelectric properties was analyzed. The elastic

The relevance of elastic constants and piezoelectric proper ties of β-$(Al_xGa_{1-x})_2O_3$ was examined in regards of strain management and mechanical stability for HEMT applications. In particular, the quantitative relationship between Al composition and the material's mechanical and piezoelectric properties was analyzed.

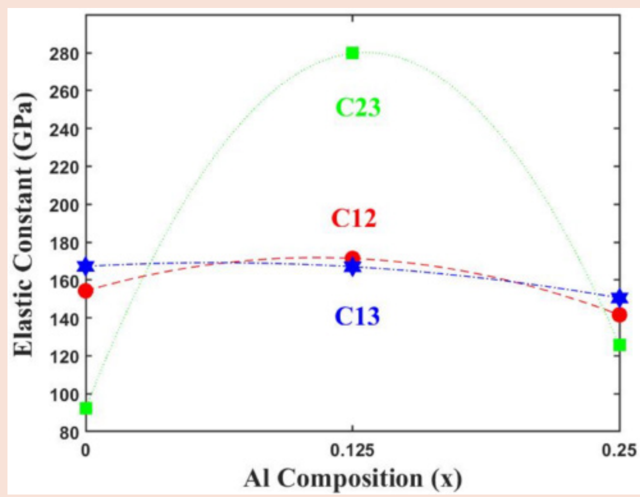

FIGURE 4 In-plane stress interaction (C_{12}, C_{13}, C_{23}).

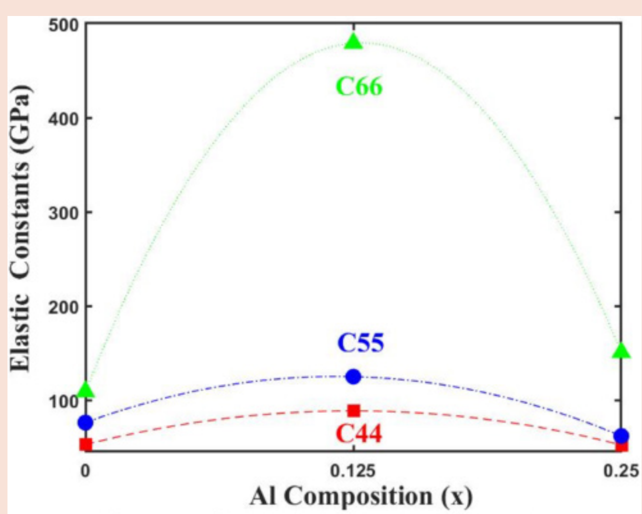

FIGURE 5 Shear strain resistance (C_{44}, C_{55}, C_{66}).

The calculated elastic and piezoelectric properties of β-$(Al_xGa_{1-x})_2O_3$ demonstrate its suitability for HEMT heterostructure designing. The increase in C_{11} and the reduction in lateral elasticity (C_{13}) with Al doping enhance mechanical stability and support strain management in device operation. Additionally, the increasingly negative d_{31} values indicates stronger strain-induced electric fields, which can improve carrier mobility and electron confinement in the channel.

TABLE 4 Piezoelectric strain coefficients and elastic constants for β-$(Al_xGa_{1-x})_2O_3$.

COMPOSITION (x)	x = 0.0	x = 0.125	x = 0.25	x = 0.0625 REF.[21]
C_{11} (GPa)	246.7	283.0	295.6	246.98
C_{13} (GPa)	167.2	167.0	150.5	167.21
e_{11} (C/m^2)	0.78×10^{-10}	0.78×10^{-10}	0.78×10^{-10}	0.78×10^{-10}
e_{31} (C/m^2)	-5.51×10^{-10}	-5.51×10^{-10}	-5.51×10^{-10}	-5.51×10^{-10}
d_{11} (C/N)	3.16×10^{-13}	2.76×10^{-13}	2.64×10^{-13}	3.16×10^{-13}
d_{31} (C/N)	-3.30×10^{-12}	-3.30×10^{-12}	-3.66×10^{-12}	-3.30×10^{-12}

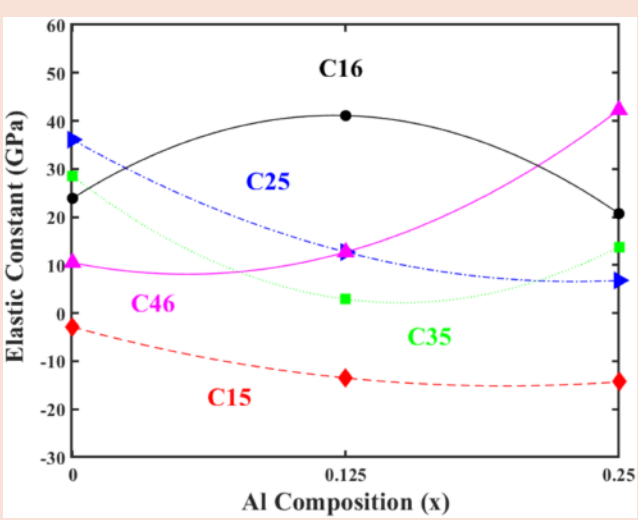

FIGURE 6 Asymmetric shear strain (C_{16}, C_{25}, C_{46}, C_{35}, C_{15}).

constants C_{ij} and piezoelectric coefficients e_{ij}, d_{ij} for the β-$(Al_xGa_{1-x})_2O_3$ alloy were evaluated by interpolation within $x \le 0.25$ from the three DFT values in Table 3 ($x = 0, 0.125$ and 0.25). For $x \approx 0.0625$ values were obtained by piecewise-linear interpolation between $x = 0$ and $x = 0.125$, and the results were compared with an experimental data at a fixed aluminum composition of 6.25% ($x = 0.0625$) [21]. At $x = 0.0625$, the elastic constants were calculated as $C_{11} = 246.98\,GPa$ and $C_{13} = 167.21\,GPa$. Using the reported piezoelectric stress coefficient ($e_{11} = 0.78 \times 10^{-10} C/m^2$, $e_{31} = -5.51 \times 10^{-10} C/m^2$) [21], the piezoelectric strain coefficients were computed as $d_{11} = 3.16 \times 10^{-13} C/N$, $d_{31} = -3.30 \times 10^{-12} C/N$. These results are consistent with the reference data, supporting the internal consistency of our DFT dataset and the interpolation.

When the fitting equations were further applied to higher Al compositions of $x = 0.125$, $x = 0.25$, the trend shows increasing C_{11} and decreasing as summarized in Table 4. These changes reflect increased stiffness and reduced lateral elasticity with the increased Al doping, whichinfluencethepiezoelectricresponse. Specifically, d_{31} becomes more negative as C_{13} decreases, indicating a stronger piezoelectric field effect under strain.

The calculated elastic and piezoelectric properties of β-$(Al_xGa_{1-x})_2O_3$ demonstrate its suitability for HEMT heterostructure designing. The increase in C_{11} and the reduction in lateral elasticity (C_{13}) with Al doping enhance mechanical stability and support strain management in device operation. Additionally, the increasingly negative d_{31} values indicates stronger strain-induced electric fields, which can improve carrier mobility and electron confinement in the channel. The polarization effects at the heterojunction interface, driven by the piezoelectric properties of β-$(Al_xGa_{1-x})_2O_3$, lead to the formation of a two-dimensional electron gas (2DEG). As the Al composition increases, the enhanced polarization effects result in higher electron concentrations in the 2DEG, further improving the device's performance [22]. These findings provide a clear pathway for β-$(Al_xGa_{1-x})_2O_3$ based heterostructures in high-performance HEMT devices. By modeling the compositional dependency of elastic and piezoelectric properties, this study offers insights into tailoring material characteristics for strain-engineered heterojunctions in HEMT applications.

IV. CONCLUSION

This study conducted a detailed investigation into the structural stability and elastic properties of β-$(Al_xGa_{1-x})_2O_3$ across various aluminum compositions (x = 0, 0.125, 0.25). Our findings are essential because β-Ga_2O_3, though promising for high-power electronics, has limitations such as poor thermal management and doping inefficiency. Aluminum gallium oxide has the potential to address these shortcomings by modifying lattice parameters and improving stability and performance through adjustable aluminum content. Understanding the elastic properties and how they evolve with Al composition provides critical insights into how Al incorporation affects device reliability and performance.

Furthermore, the investigation of 13 elastic constants, characteristic of the monoclinic structure, and their fitted polynomial equations (linear for density, cubic for volume and β allows for precise modeling of mechanical properties as a function of aluminum content. This is crucial for predicting material behavior in high-power and optoelectronic devices, where stability under stress and lattice matching are vital. The fitted equations, particularly for constants such as C_{11} and C_{44}, provide predictive capabilities to optimize material compositions for future applications. As such, this work serves as a foundation for further exploration of β-$(Al_xGa_{1-x})_2O_3$, guiding the development of advanced materials for semiconductor technologies.

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